

Nguyen Khac Dong Quan

Backend Developer Intern

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OBJECTIVE

[Position] with experience developing [type of systems/projects] using [core stack]. Short-term goal is to improve [specific technical area] and transition into a [target role] within [timeframe]. Seeking an opportunity at [Company Name] to apply technical skills, learn from experienced developers, and contribute effectively to real-world software projects.

SKILLS

Programming Languages: Java, JavaScript (Node.js)

Core Concepts: RESTful APIs, JWT, OAuth2, Socket.IO, Data Structures & Algorithms

Backend Frameworks: Spring Boot, Spring Security, Spring Data JPA, Express.js, Spring AI

Databases & Persistence: MySQL, MariaDB, Qdrant (Vector Database), Hibernate, MapStruct

Software Architecture: Layered Architecture, Modular Monolith, Monolithic Architecture

Tools & AI Productivity: Docker, Git, Linux, Github Copilot, Gemini CLI, Context/Token Management, Prompt Engineering

PROJECTS

Perfume E-commerce | Team size: 1 | 🐙 [GitHub Repository](#) | 📺 [Video Demo](#)

February 2026 – May 2026

Technologies: Java 21, Spring Boot 3, Spring AI, Qdrant (Vector DB), OAuth2/JWT, Docker, VNPay

- Built a **RAG-based AI consultant** with **Spring AI** and **Qdrant**, reducing manual support overhead by **70%** through 24/7 automated expert-level perfume advice.
- Integrated **VNPay v2.1** to provide a **seamless and secure cashless payment** experience, including a **Repay flow** to ensure successful transactions even after session timeouts.
- Integrated **Cloudinary** for automated asset management, offloading server storage and ensuring fast image delivery via CDN-based transformations.
- Designed a **modular monolith** architecture, accelerating development cycles for new features by **30%** through decoupled domains and MapStruct mapping.
- Optimized persistence with **EntityGraph** and **BatchSize**, reducing database queries **from N to 1-2** and improving response times by **90%** for heavy product listings.
- Architected **JWT + OAuth2 account linking** with SEO-friendly public access, streamlining the authentication flow and increasing user conversion rates.
- Built **async email notifications** via Spring Boot Mail, replacing manual invoicing labor and ensuring **100%** delivery of instant order confirmations.

POS | Team size: 1 | 🐙 [GitHub Repository](#) | 📺 [Video Demo](#)

May 2025 – August 2025

Technologies: Node.js, Express.js, MySQL, Socket.IO, EJS, RESTful APIs, express-session, bcrypt, Docker

- Engineered a **multi-directional real-time** system using **Socket.IO** with **Rooms/Namespaces**, isolating data by table/role and reducing **turnover time by 20%**.
- Architected instant event flows between **Customers** (order status), **Kitchen** (new orders), and **Staff** (support/payment requests) without page reloads.
- Designed a robust **order state machine** (Pending → Cooking → Served), maintaining strict data consistency across 3 distinct user roles.
- Ensured **data integrity** under high concurrency using **MySQL transactions**, preventing duplicate billing or lost orders during peak restaurant hours.
- Implemented a **hybrid SSR/API** architecture, providing fast initial loads for administrative dashboards while supporting dynamic client-side updates.
- Secured sensitive financial records through **RBAC middleware** and **Bcrypt** hashing, ensuring **0%** unauthorized access to revenue data.
- Developed an automated **reporting engine** using SQL aggregations, providing instant daily revenue analytics and customer loyalty tracking.

EXPERIENCE

Nhat Minh Education Center <i>Math Tutor</i>	February 2025 – August 2025 <i>Ho Chi Minh City, Vietnam</i>
- Successfully mentored 10+ students, significantly boosting academic results: 3 students jumped from failing grades (3-4) to excellence (9+), while others improved from 5-6 to 8+. - Tailored teaching methods to individual learning styles, demonstrating strong patience, communication, and result-oriented mentorship.	
Freelance Web Development <i>Frontend Developer</i>	January 2024 – Present <i>Ho Chi Minh City, Vietnam</i>
- Developed 5+ customized static web solutions using HTML, CSS, and JavaScript based on specific academic project requirements for university students. - Ensured 100% adherence to technical specifications and design handoffs, delivering clean and maintainable code for various educational modules.	
Computer Repair Services <i>Freelance Technician</i>	January 2023 – Present <i>Ho Chi Minh City, Vietnam</i>
- Resolved 20+ hardware and software cases with high client satisfaction; performed Windows installations, RAM upgrades, and system troubleshooting. - Built a reputation for reliable technical support and quick problem diagnosis for individual clients in the local area.	

EDUCATION

Sai Gon University <i>Bachelor of Engineering in Information Technology</i>	September 2022 – March 2027 <i>Current GPA: 7.8/10</i>
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CERTIFICATES

Aptis ESOL B2 <i>English Proficiency Certificate</i>	March 2026 <i>British Council (BC)</i>
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HOBBIES

Reading Books
Regularly read books about psychology and communication to improve interpersonal and analytical thinking skills.
Playing Piano
Practice piano during free time as a hobby to relax and improve focus and discipline.